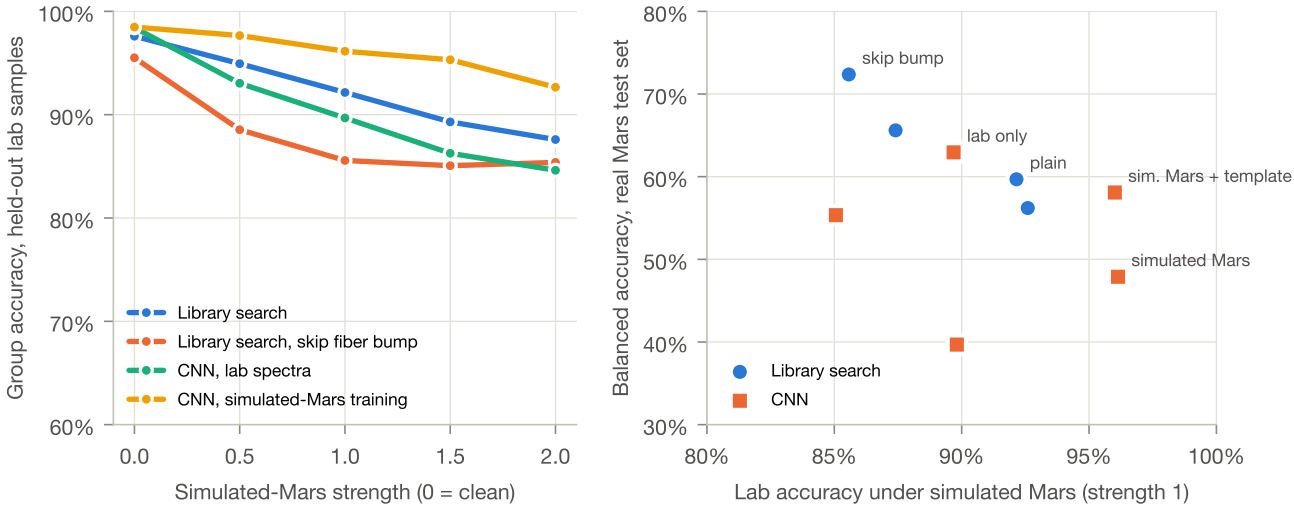
(a) Robustness to simulated Mars
100%
90%
80%
70%
60%
Group accuracy, held-out lab samples
0.0
0.5
1.0
1.5
2.0
Simulated-Mars strength (0 = clean)
Library search
Library search, skip fiber bump
CNN, lab spectra
CNN, simulated-Mars training
(b) Simulated robustness vs. real Mars
80%
70%
60%
50%
40%
30%
Balanced accuracy, real Mars test set
80%
85%
90%
95%
100%
Lab accuracy under simulated Mars (strength 1)
skip bump
lab only
plain
sim. Mars + template
simulated Mars
Library search
CNN